Chapter 12

Cost Index & Multi-Asset Trading Costs

INTRODUCTION

This chapter is separated into two sections. In the first section we introduce readers to the cost index methodology. In the second section we introduce techniques for investing and trading across multi-asset classes.

First, cost indexes provide investors with trading cost estimates based on actual market conditions and aggregated buying and selling pressure across all market participants. We derive cost indexes for historical and real-time data sets. These assist investors to improve stock selection, back-test investment ideas, perform portfolio attribution analysis, critique algorithms, and hold their providers accountable for performance in real-time. We also present a market impact simulation experiment that can be used to evaluate and construct stock specific market impact models. This simulation technique shows the difficulty in formulating these models and estimating corresponding parameters at the stock level even when the true parameters are known. The cost index has become an essential (if not the most essential) trading cost metric for investors.

We next turn our attention to investing methodologies across different asset classes. We discuss the trading caveats of trading similar instruments such as equities, exchange traded funds, and futures. For example, many times a portfolio manager's investment objective is to acquire a specified beta exposure to the market or SP500 index. Other times a manager's objective is to acquire exposure to a different factor or macro economic indicator. This is commonly referred to as "Investing in Beta" or "Investing in Factor Exposure." The manager determines what the end goal is but is indifferent as to which underlying instruments are used to acquire that exposure. We provide techniques to determine how to best acquire the specified exposure given market conditions and trading costs. Managers could improve portfolio performance through more efficient implementation. As we show, many times the optimal portfolio mix will be an allocation across the three asset types: equities, exchange traded funds, and futures. The chapter concludes with the introduction of an

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